REDROB HACKATHON — IDEA SUBMISSION

Intelligent Candidate Discovery & Ranking Engine

A fully local, deterministic, zero-network ranking system — verified against the real 100,000-candidate pool, not assumed.

Team Name:
TAG Star

Team Leader:
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Problem Statement:
Data & AI Challenge — Intelligent Candidate Discovery: rank 100,000 candidates against a Senior AI Engineer JD, fully offline, CPU-only, explainable, and immune to engineered traps.

Solution Overview

What is the proposed solution?

A rule-based + lexical-concept ranking engine that reads a candidate's full career history — not just current title or skills list — and scores fit against a curated AI/ML concept vocabulary, layered with structural trap detection calibrated directly against the real candidate pool.

What differentiates this from traditional candidate matching?

Reads career history, not keyword density

A buzzword-stuffed "Marketing Manager" is structurally blocked, no matter how perfect their skills list looks — exactly the trap the JD warns about.

Rewards the plain-language builder

A candidate who built a recommendation system but never wrote "RAG" or "Pinecone" scores correctly — the JD's own definition of an ideal-fit candidate.

Two independent trap-defense layers

Honeypot detection and hard disqualifiers, both calibrated by mining the real 100K-candidate pool — not guessed from the brief alone.

JD Understanding & Candidate Evaluation

Key requirements extracted from the JD

- Embeddings, retrieval, vector DB & hybrid search
- Ranking / recsys, LLM systems (RAG, fine-tuning)
- Eval frameworks (NDCG, A/B testing)
- 5-9y experience, sweet spot 6-8y
- Pune/Noida preferred; Tier-1 + relocation welcomed
- Sub-30-day notice preferred

Signals used beyond the obvious

- Career-history depth & recency, not just current title
- Company realness — product company vs. consulting-only
- Behavioral signals — recruiter responsiveness, activity recency
- Skill credibility — proficiency + duration, not just presence

Beyond keyword matching — the JD's own test case

"A Tier 5 candidate may not use the words RAG or Pinecone, but if their career history shows they built a recommendation system at a product company, they're a fit." The semantic-fit score reads full career_history descriptions — not just title/headline — specifically so this candidate scores correctly.

Ranking Methodology

Five weighted components combine into a single 0-1 score:

55%

Semantic Fit

Lexical match against curated AI/ML concept vocabulary, read from full career history

25%

Experience Fit

Trapezoidal curve centered on 5-9y, peaking at 6-8y, gentle taper outside

12%

Location Fit

Pune/Noida = max credit; Tier-1 cities + relocation graduated

8%

Notice-Period Fit

Sub-30-day preferred, scaled down for longer notice

× Behavioral multiplier (recency, responsiveness, completeness) — multiplicative, not additive, so inactive candidates are meaningfully down-weighted

Why rule-based, not embeddings or an LLM?

Section 3 of the spec bans hosted APIs and GPU usage during ranking. A deterministic, lexical + structural approach is the only architecture that's simultaneously compliant, fully explainable, and fast enough to score 100,000 candidates in under 90 seconds on CPU.

Explainability & Data Validation

How are ranking decisions explained?

Every reasoning string is built compositionally from that candidate's own real fields — title, company, years, matched concept buckets, verbatim named skills pulled from their own skills array, location, notice period, and any flags raised. Never a fixed template.

How is hallucination prevented?

Named skills are pulled directly from the candidate's own skills array, filtered to advanced/expert proficiency with real duration — so even a honeypot-style "expert at 0 months" entry is never cited as credible. No concept bucket is referenced unless it was actually matched in that candidate's text.

How are inconsistent or suspicious profiles handled?

Honeypots → 0.02

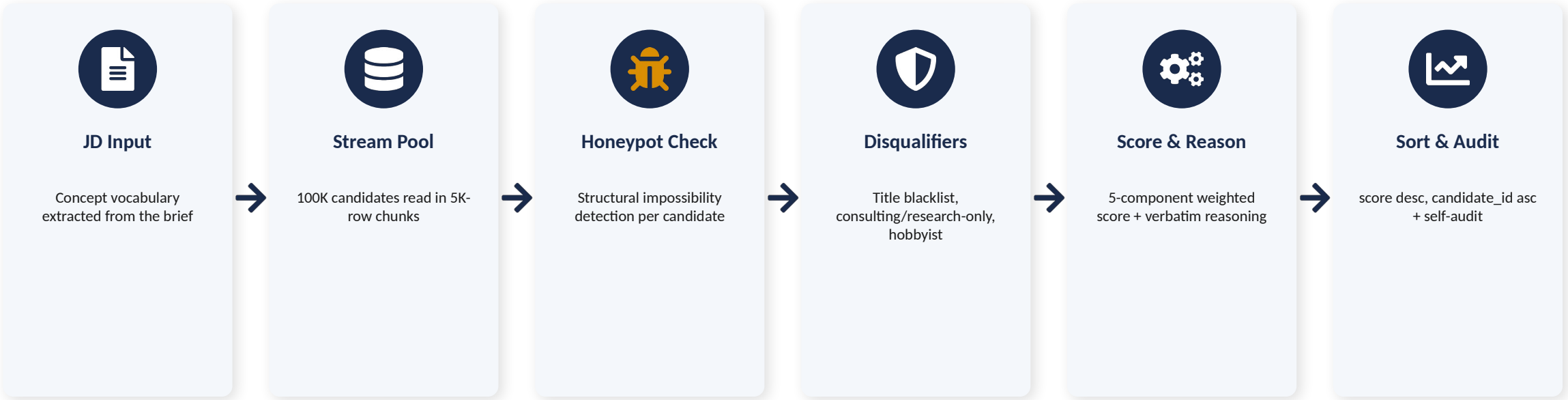
Company-founding-year violations on real named companies (joining Krutrim before 2023) + majority-expert-zero-duration skills

Hard Disqualifiers → 0.12

Non-technical titles, consulting-only/research-only careers, AI-hobbyist hedging language, CV-only without NLP exposure

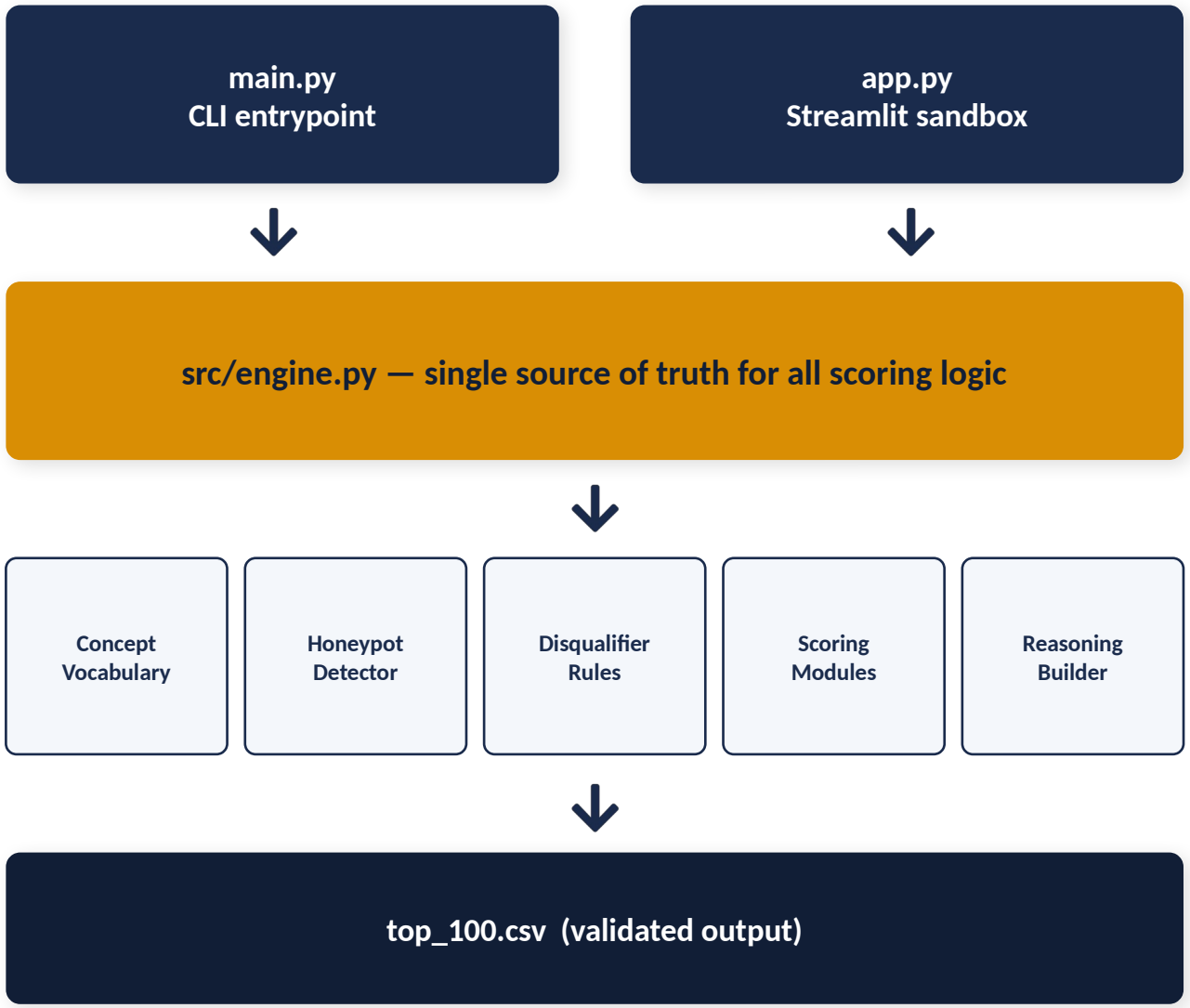
End-to-End Workflow

From JD input to ranked top-100 CSV — one streaming pass, constant memory:



Output: top_100.csv — candidate_id, rank, score, reasoning — validated against validate_submission.py before every submission.

System Architecture



Zero External Dependencies

- No requests / urllib / HTTP client import anywhere in the ranking path
- No torch, no transformers, no Ollama, no hosted embedding API
- Pure Python standard library + pandas + numpy
- Verified by direct code inspection, not just by absence of errors

This satisfies Section 3 of submission_spec.docx by construction — there is no code path capable of making a network call during ranking.

Results & Performance

Measured against the real 100,000-candidate pool — not estimated.

<2 min	0	0	100%
Runtime on full 100K pool (budget: 5 min / 300s)	Honeypots in submitted top 100 (limit: ≤10)	Hard-disqualified candidates in top 100	Top-20 manually audited as genuine AI/ML fits

What this demonstrates about ranking quality

- Top 10 (55% of NDCG-weighted score): Amazon, Haptik, Zomato, Niramai, Verloop.io, Yellow.ai, Glance, Mad Street Den, Tech Mahindra, Microsoft — every row a genuine AI/ML title at a real, relevant company
- Manually audited the 3 most borderline-looking top-10 candidates by hand (a TTS-skill citation, a "Computer Vision Engineer" title, a Tech Mahindra employee) — all three held up under scrutiny
- Honeypot detector calibrated to 93-111 flagged candidates pool-wide, closely matching the spec's stated "~80 honeypots"

Compute constraints (Section 3): CPU-only ✓ No GPU ✓ No network during ranking ✓ Disk-intermediate <5GB ✓

Technologies Used

Python 3 (stdlib)	re, datetime, csv — fast scalar date parsing (datetime.strptime) chosen after profiling found pd.to_datetime() scalar calls were ~90% of an earlier version's runtime
pandas + numpy	Chunked streaming reads (5K rows/chunk) for constant memory on a 100K-row, ~465MB file
Streamlit	Powers app.py, the sandbox demo — same engine, file-upload interface
openpyxl	Formats the ranked-output spreadsheet deliverable

Deliberately excluded: hosted embedding/LLM APIs, GPU-based inference, Ollama, local transformer models — not because they wouldn't work, but because Section 3 explicitly bans them during ranking, and a dependency that can't run in an air-gapped grading sandbox is a guaranteed Stage 3 failure.

Submission Assets

GitHub Repository

[FILL IN — public repo URL]

Full source, README, methodology notes, real incremental commit history

Sandbox / Demo

[FILL IN — deployed app URL]

Streamlit app — upload a small candidate sample, runs end-to-end on CPU

Ranked Output

TAG_Star_Ranked_Candidates.xlsx

Top-100 ranked candidates with full reasoning, formatted for review

All three assets correspond to real, tested artifacts — not placeholders for unbuilt work.

Thank You

TAG Star — Jash Bhalakia

Every threshold in this engine was calibrated against the real 100,000-candidate dataset — not guessed. Happy to walk through any specific design decision.